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Association between Attention Deficit Hyperactivity Disorder and Obstructive Sleep Apnea in Children: A Systematic Review and Meta-Analysis

Bryan Hao Wei Leow[1], Claire Jing-Wen Tan[1], Brian Sheng Yep Yeo[2], Noelle Na En Ng[2], Seraphina Cheng Yong Low[2], Adele Chin Wei Ng[1,3,4], Oon Hoe Teoh[3,5,4], Song Tar Toh[1,3,4], and Ellen Ghim Hoon Tay[4,5]

Abstract

Background: Obstructive Sleep Apnea (OSA) and Attention Deficit Hyperactivity Disorder (ADHD) are two relatively common conditions in the pediatric population, with multiple studies demonstrating a strong association between the two. Hypoxia and sleep fragmentation caused by OSA has been shown to exacerbate ADHD symptoms, while treatment for OSA, such as with continuous positive airway pressure or adenotonsillectomy, may improve ADHD symptoms. We aim to provide an updated systematic review and meta-analysis on the complex relationship between OSA and ADHD in children.
Methods: PubMed, Embase, and SCOPUS were searched from inception to 1 December 2024 for longitudinal studies investigating the association between OSA and ADHD in children. Two independent authors selected relevant articles, extracted data, assessed bias using the Newcastle-Ottawa Scale and evaluated quality of evidence following the Grading of Recommendations, Assessment, Development and Evaluations framework. Random-effects meta-analysis was performed to synthesize pooled prevalence, while descriptive reviews were performed for all other outcomes.
Results: This systematic review and meta-analysis of 11 studies and 903 children formally diagnosed with OSA or ADHD found that the pooled prevalence of OSA in children with ADHD was 44% (95% CI [2%, 56%]). Descriptive analysis found that adenotonsillectomy for OSA consistently led to significant improvements in both OSA and ADHD symptoms, suggesting its potential in complementing existing pharmacological therapy in improving ADHD symptoms.
Conclusion: OSA and ADHD are closely related comorbid conditions. Physicians should be cognizant of this association and consider concurrent evaluation for OSA in children with ADHD symptoms, which can effectively guide treatment strategies and potentially reduce the need for long-term pharmacological treatment. (*J. of Att. Dis. 2026; 30(6) 822-832*)

Keywords

obstructive sleep apnea, attention deficit hyperactivity disorder, inattention, hyperactivity, adenotonsillectomy, children, pediatrics

Introduction

Attention Deficit Hyperactivity Disorder (ADHD) is a neurodevelopmental disorder characterized by patterns of inattention, hyperactivity, and impulsivity that interfere with daily functioning or development. It is usually diagnosed in childhood, though it may persist into adolescence and adulthood. ADHD is diagnosed clinically in accordance with the Diagnostic and Statistical Manual for Mental Disorders-5 (DSM-5; Substance Abuse and Mental Health Services Administration, 2016).

Obstructive Sleep Apnea (OSA) is a form of sleep-related disordered breathing characterized by recurrent episodes of

[1]Singapore General Hospital, Singapore
[2]National University of Singapore, Singapore
[3]SingHealth Duke-NUS Sleep Centre, Singapore
[4]Duke-NUS Medical School, Singapore
[5]KK Women's and Children's Hospital, Singapore

Corresponding Authors:
Ellen Ghim Hoon Tay, Department of Child Development, KK Women's and Children's Hospital, 100 Bukit Timah Road, Singapore 229899, Singapore.
Email: ellen.tay.g.h@singhealth.com.sg

Bryan Hao Wei Leow, Department of Otorhinolaryngology - Head and Neck Surgery, Singapore General Hospital, Singapore 168753, Singapore.
Email: bryan.leow@mohh.com.sg

partial or complete obstruction of the upper airway during sleep, leading to intermittent cessation of airflow and sleep arousal. The underlying causes of OSA in children can be multifactorial, with major risk factors being adenotonsillar hypertrophy and obesity, and other risk factors being prematurity, asthma, allergic rhinitis, craniofacial abnormalities, neuromuscular disorders and a family history of OSA (Hasuneh et al., 2024). Clinically, OSA severity is categorized by the Apnea-Hypopnea Index (AHI). It can be challenging to diagnose OSA in children as presentation may be distinct from that in adults. Children with OSA usually present with snoring, night awakenings, unusual sleep positions and sometimes nocturnal enuresis. In children, the sleep disruption may also lead to behavioral issues including hyperactivity, irritability, and even aggression. Adenotonsillectomy is the most common treatment for pediatric OSA (American Academy of Pediatrics. Section on Pediatric Pulmonology, Subcommittee on Obstructive Sleep Apnea Syndrome, 2002).

Several studies have shown that OSA and ADHD in children may be closely linked. Chronic sleep disruption caused by OSA can exacerbate the core symptoms of ADHD, leading to a cycle of worsening sleep disturbances and behavioral difficulties. Conversely, the hyperactive and impulsive behaviors commonly associated with ADHD may mask the presence of OSA in affected children, and lead to complications such as developmental delays, behavioral issues, cardiovascular problems, and impaired cognitive function (Youssef et al., 2011).

Of note, a previous systematic review and meta-analysis by Urbano et al. found that adenotonsillectomy in children with OSA was effective in the short-term treatment of ADHD symptoms. However, the success of other treatment options including Continuous Positive Airway Pressure (CPAP) therapy was not established (Urbano et al., 2021). Another meta-analysis by Sedky et al. demonstrated that ADHD symptoms are related to OSA and improve after adenotonsillectomy, reducing the need for reliance on pharmacological intervention for control of ADHD. As such, it would be worthwhile to examine the association between these two comorbid conditions to guide clinical management (Sedky et al., 2014).

We aim to perform an updated systematic review and meta-analysis to summarize the latest literature and provide deeper insights on the relationship between OSA and ADHD in children.

Materials and Methods

According to a pre-registered protocol on PROSPERO (CRD42025645881), this systematic review and meta-analysis was conducted according to Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) recommendations (Page et al., 2021). The PRISMA checklist can be found in Supplemental Table S1.

Search Strategy

PubMed, Embase and Cochrane Library were searched from inception to 1 December 2024, using search terms pertaining to OSA and ADHD, treatment methods and their relevant synonyms.

Study Selection

The selection of eligible articles by titles and abstracts, followed by full texts, was performed by three independent authors (CJWT, NNEN, SCYL), with disagreements resolved through the consensus of another independent author (BHWL). Based on the Population, Intervention, Comparators, Outcomes (PICO) framework, we selected studies that met the following inclusion criteria: randomized controlled trials or observational studies (cohort, cross-sectional) published as full-length articles in peer-reviewed journals studying the associations between OSA and ADHD in children. We included peer-reviewed observational studies that were published as full-length articles in English and utilized formal methods of diagnosis of OSA and ADHD. We utilized an AHI cut-off of $\geq$1 event per hour for the diagnosis of pediatric OSA, in accordance with the American Academy of Pediatrics guidelines (Marcus et al., 2012). We excluded studies that did not meet the inclusion criteria, duplicate studies, case reports, case series, reviews, meta-analyses, letters, conference abstracts, and studies with incomplete data.

Data Extraction

We extracted data from each included article into a standardized extraction template. The data extraction process was performed by two independent authors (CJWT, BHWL). We extracted the following information: first author, year of publication, study design, country, follow-up duration, total number of participants, age, gender, prevalence of OSA in ADHD patients and vice versa, severity of OSA by AHI, intervention for OSA and ADHD respectively, subjective and objective outcomes following treatment, covariates, statistical methods and key findings.

Risk of Bias Assessment

The risk of bias of observational studies was appraised by four independent authors (CJWT, BHWL, NC, SL) using the Newcastle-Ottawa Scale (NOS; Wells et al., 2014) acknowledged by the Cochrane Collaboration (Higgins et al., 2017), for interventional and observational studies respectively. The NOS is a nine-point grading system derived from an eight-item checklist, designed to evaluate the extent of selection, information, and confounding biases, in determining the risk of bias in non-randomized studies. As per the NOS grading in past reviews, we graded the studies as having a high ($<$5 stars), moderate (5–7 stars)

or low risk of bias (≥8 stars; Wells et al., 2014). Risk of bias assessment via the Newcastle-Ottawa scale is demonstrated in Supplemental Table S2.

Statistical Analysis

We conducted all statistical analyses on RevMan (version 5.4), in accordance with statistical approaches laid out by the Cochrane handbook (Higgins et al., 2017). Unless otherwise specified, a p-value of ≤.05 was considered statistically significant.

When two or more studies were available, we proceeded with our planned meta-analyzes. Using a random-effects model, maximally covariate-adjusted ratios were pooled from each study. We assessed for between-study heterogeneity using Thompson's I^2 values and the Cochran Q Test, where an I^2 value of 25%, 50% and 75% represented low, moderate, and high degree of heterogeneity respectively and a τ^2 p-value of ≤.10 was considered significant for heterogeneity. To assess for publication bias, we qualitatively examined funnel plots for asymmetry and imputed potentially missing studies using the trim-and-fill method.

For all other outcomes in which there were insufficient data for a meta-analysis, descriptive analysis was performed.

Results

The PRISMA flowchart demonstrating the study selection process is presented in Figure 1. Literature search of the 4 databases (PubMed, Embase, Scopus and Cochrane Library) retrieved 825 results. 292 duplicates were removed. Title and abstract screening further excluded 498 articles. Full text screening excluded 24 articles. 11 articles were included in the final analysis (Chin et al., 2018; Dillon et al., 2007; Galland et al., 2011; Goraya et al., 2009; Huang et al., 2004, 2007; Li et al., 2006; Nguyen-Ngoc-Quynh et al., 2022; Pagel et al., 2004; Tsai et al., 2012; Wu et al., 2023).

Study Characteristics

Baseline study characteristics of the included studies are summarized in Table 1. Briefly, of the 11 included studies, eight cohort studies, two case-control studies and one cross-sectional study. The proportion of males in the included studies ranged from 51.9% to 90%, while mean age ranged from 7.8 to 10.7 years. Six studies had low risk of bias while five studies had moderate risk of bias as per the Newcastle-Ottawa Scale in Supplemental Table S2.

Meta-Analysis

Four studies reported the prevalence of OSA in ADHD patients (Goraya et al., 2009; Huang et al., 2004; Pagel et al., 2004; Tsai et al., 2012). The pooled prevalence of OSA in children with ADHD was 44% (95% CI [32%, 56%], I^2 = 72.2%; Figure 2).

Systematic Review

Association Between OSA and ADHD. Two studies investigated the prevalence of ADHD in children with OSA. Nguyen et al. found that the prevalence of ADHD in children with OSA is 43.1% (25 events out of 58 subjects), while Dillon et al. found that the same prevalence is 27.8% (22 events out of 79 subjects; Dillon et al., 2007; Nguyen-Ngoc-Quynh et al., 2022).

Three studies investigated the association between AHI and ADHD. Wu et al. demonstrated that attention deficit and hyperactive impulsivity scores graded by an 18-item ADHD rating scale were independently associated with the obstructive AHI ($p < .001$; Wu et al., 2023). Similarly, Chin et al. found that children with ADHD had significantly higher AHI ($p = .002$), hypopnea counts ($p = .003$) and lower slow-wave sleep percentages ($p = .006$) on polysomnography (Chin et al., 2018). Conversely, categorical analyses performed by Galland et al. in a small study population did not show any significant difference in AHI between children with and without ADHD (Galland et al., 2011).

Adenotonsillectomy

ADHD Parameters. Dillon et al. demonstrated that following adenotonsillectomy for the clinical indication of obstructed breathing while awake, obstructed breathing while asleep and recurrent tonsillitis, the overall prevalence of ADHD declined modestly from 27.8% to 20.5%. Among the 22 children diagnosed with ADHD at baseline, 50% remitted and no longer fulfilled DSM-IV criteria for ADHD, while the remaining 50% continued to have ADHD. The follow up prevalence remained high at 20.5% despite the 50% remission of ADHD because five children were newly diagnosed with ADHD at follow up (Dillon et al., 2007).

Two studies examined changes after adenotonsillectomy on the Disruptive Behavior Disorder Rating cale (DBDRS). Dillon et al. found that among the children whose ADHD remitted after adenotonsillectomy, rating changes on DBDRS corresponding to preoperative diagnosis changed by an average of 51.1%, with a mean decrease in DBDRS inattentive, hyperactivity and combined scores by 8.4, 6.0 and 18.0 respectively (Dillon et al., 2007). Huang et al. (2007) also demonstrated statistically significant changes in the overall DBDRS scores and hyperactivity/impulsivity sub scores post-adenotonsillectomy ($p = .007$ and $p = .002$ respectively; Huang et al., 2007).

Similarly, two studies utilized the Child Behavior Checklist (CBCL). Huang et al. (2007) found that multiple parameters in the CBCL rating scale showed statistically significant changes post-adenotonsillectomy, namely for 'thought/obsessive', 'somatic complaint', 'social withdraw',

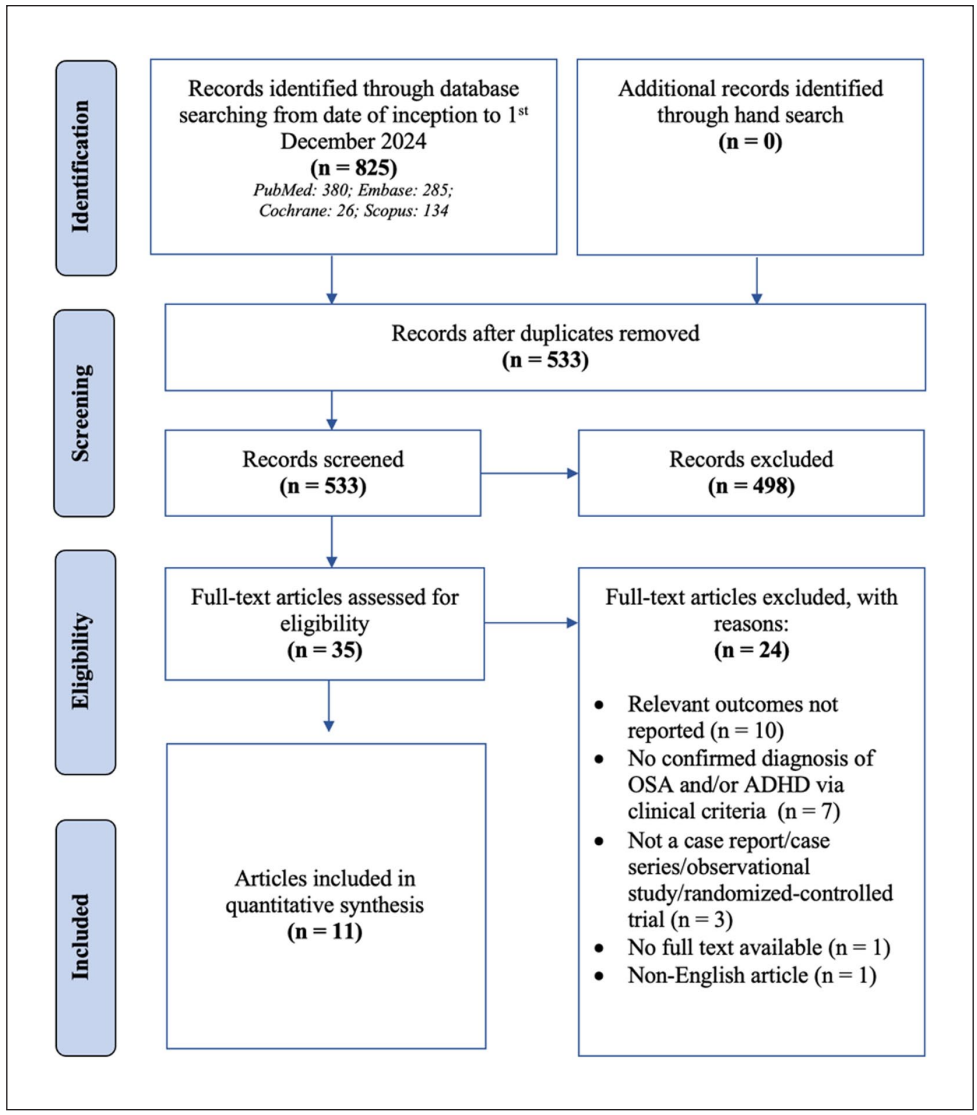


Figure 1. PRISMA flow chart.

'delinquent' and 'internalizing' ($p < .05$; Huang et al., 2007). Similarly, in different study, Li et al. found that all parameters in the CBCL rating scale showed significant changes post-adenotonsillectomy, with greatest change in 'depression/anxiety', 'thought/obsessive' and 'internalizing' ($p = .001$; Li et al., 2006).

The same two studies also utilized the Test of Variables of Attention (TOVA). Both Huang et al. (2007) and Li et al. found that the change in response time post-adenotonsillectomy were statistically significant, from -0.83 ± 1.22 to -0.32 ± 0.94 and -0.79 ± 1.14 to -0.28 ± 0.93 respectively ($p = .0006$ for both; Huang et al., 2007; Li et al., 2006). Overall ADHD scores post-adenotonsillectomy also showed statistically significant changes in both studies ($p < .001$).

Lastly, Huang et al. (2007) also analyzed changes in the ADHD rating scale (ADHD-RS) and found that children treated with adenotonsillectomy for OSA had an ADHD total score of 21.16 ± 7.13 on the ADHD-RS post-surgery compared to 31.52 ± 7.01 pre-surgery ($p = .0001$). Moreover, the attention and hyperactivity subscales were also significantly lower ($p < .0001$) in this group of children post-operatively (Huang et al., 2007).

Sleep Parameters. Two studies examined changes in sleep parameters following adenotonsillectomy for OSA.

Two studies evaluated changes in PSG parameters. Huang et al. (2007) demonstrated statistically significant changes to AHI following adenotonsillectomy in children diagnosed with ADHD, from pre-surgery AHI of 3.32 ± 1.11 to post-surgery AHI of 0.89 ± 0.63 ($p < .001$). The same study also noted statistically significant changes in snoring index, arousal index and total sleep time ($p < .05$ for all

Table 1. Baseline Study Characteristics of Included Studies.

Study	Country	Study design	Mean follow-up duration (months)	Total participants	No. of OSA participants	No. of ADHD participants	Mean age (years)	Mean BMI	% of males	Treatment/intervention studied
Nguyen-Ngoc-Quynh et al. (2022)	Vietnam	Cross sectional study	12	96	58	32	8.4 ± 2.4	17.1 ± 2.2	60.4	Clinical characteristics of patients with OSA and ADHD
Wu et al. (2023)	China	Prospective observational study	3	228	133	NA	7.8 ± 1.6	17.1 ± 4.0	62.3	Clinical characteristics of patients with OSA and ADHD
Galland et al. (2011)	New Zealand	Matched case control study	NR	56	NR, unable to tell how many participants have OSA as only mean AHI is reported	28	10.1 ± 1.8	18.7 ± 3.9	78.6	Treatment effects of methylphenidate on ADHD participants, evaluation of change in PSG parameters with use of methylphenidate
Huang et al. (2004)	Taiwan	Cross sectional case controlled study	6	102	67 of the ADHD patients had AHI >1	88	8.4 ± 1.8	18.2 ± 3.6	88.2	Clinical characteristics of patients with OSA and ADHD
Huang et al. (2007)	Taiwan	Prospective observational study	6	86	All 66 ADHD patients had AHI >1	66	8.1 ± 1.2	18.5 ± 1.2	87.2	Treatment effects of adenotonsillectomy and methylphenidate on OSA and ADHD symptoms
Goraya et al. (2009)	USA	Retrospective cohort study	12	33	8 of the ADHD participants had AHI >1	33	9.0 ± 1.4	21 ± 1.7	79.0	Clinical characteristics of patients with OSA and ADHD
Pagel et al. (2004)	Germany	Prospective observational study	NR	74	74:29 (AHI cut off >5), 45 participants had AHI <5	NR	10.7 ± 1.5	21.8 ± 2.2	58.6	Clinical characteristics of patients with OSA and ADHD
Dillon et al. (2007)	USA	Prospective cohort study	13	106	40	22	8.1 ± 1.8	NR	51.9	Treatment effects of adenotonsillectomy on OSA and ADHD symptoms
Chin et al. (2018)	Taiwan	Prospective cohort study	6	101	NR, unable to tell how many participants have OSA as only mean AHI is reported	71	8.8 ± 1.9	18.8 ± 3.9	76.0	Treatment effects of methylphenidate use on OSA and ADHD symptoms
Li et al. (2006)	Taiwan	Prospective cohort study	6	40	40:24 had SDB alone, 16 had SDB + ADHD	16	8.4 ± 1.6	18.6 ± 4.2	90.0	Treatment effects of adenotonsillectomy on OSA and ADHD symptoms
Tsai et al. (2012)	Taiwan	Case control study	12	256	2 of the ADHD participants had AHI >1	64	9.9 ± 2.5	NR	60.9	Clinical characteristics of patients with OSA and ADHD

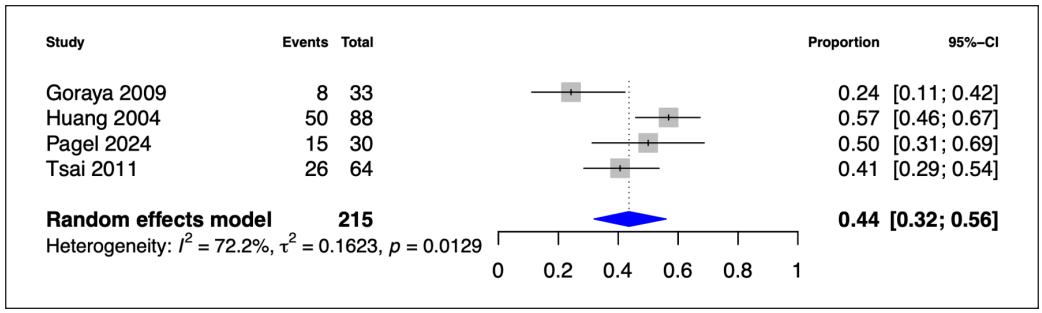


Figure 2. Prevalence of OSA in children with ADHD.
Note. Black diamonds are the estimated pooled prevalence for each random-effects meta-analysis; red box sizes reflect the relative weight apportioned to studies in the meta-analysis.

three parameters; Huang et al., 2007). In addition, Li et al. saw comparable changes to pre and post adenotonsillectomy AHI, from 10.6 ± 11.1 to 1.7 ± 2.1 ($p < .001$). However, the change in AHI did not correlate negatively with the TOVA score, suggesting that the TOVA score improvement from adenotonsillectomy may not be directly due to the reduction of sleep apnea or hypopnea events. The study also noted statistically significant changes post adenotonsillectomy for the snoring index and mean arterial oxygen saturation ($p = .014$ and $p = .018$ respectively; Li et al., 2006).

One study utilized the OSA-18 Quality of Life (QOL) rating scale in children with OSA and ADHD. Huang et al. (2007) showed statistically significant changes to sleep disturbances and overall QOL parameters following adenotonsillectomy ($p < .001$ and $p < .05$ respectively; Huang et al., 2007.

Pharmacological Treatment With Methylphenidate

ADHD Parameters. Two studies examined changes in ADHD parameters following methylphenidate use.

Huang et al. (2007) showed demonstrated a statistically significant change only in inattention sub scores ($p < .01$), but not for hyperactive sub scores or total score on the ADHD-RS after 6 months of methylphenidate use. However, there were no statistically significant changes in any of the eight subscales on the CBCL with methylphenidate use over the same time period (Huang et al., 2007).

Separately, Chin et al. found that there was a statistically significant change in both inattention and hyperactive sub scores and total scores on the ADHD-RS after 6 months of methylphenidate use in children with ADHD ($p < .05$). Findings on the CBCL were dissimilar as only externalizing behavior, anxiety, obsession/compulsion and aggression subscales showed statistically significant change with methylphenidate use ($p < .05$) while the rest of the subscales had no significant change with methylphenidate use (Chin et al., 2018).

Sleep Parameters. Two studies examined changes in sleep parameters following methylphenidate use.

In children with OSA, Huang et al. demonstrated a non-significant change in AHI from 2.24 ± 1.44 to 2.50 ± 2.95 with methylphenidate use after 6 months. In children who were not on methylphenidate, there was a change in AHI from 2.56 ± 1.46 to 2.31 ± 2.19 over the same duration. There was also no statistically significant change in other sleep parameters measured in the PSG, especially for snoring index, arousal index and total sleep time which showed significant changes post-adenotonsillectomy in the same study. Similarly, there was no significant change in OSA-18 QOL scoring for methylphenidate use, in comparison to adenotonsillectomy (Huang et al., 2007).

Similarly, Galland et al. showed that in case-matched paired analysis done between 28 pairs of ADHD children and controls, only total sleep time showed statistically significant change on the medication night in children with ADHD ($p < .01$),and showed that methylphenidate use had no significant effect on AHI, desaturation events or snoring percentage (Galland et al., 2011).

Chin et al. found a statistically significant increase in AHI from a mean of 1.20 ± 1.22 from 1.98 ± 2.78 ($p = .012$) following treatment with methylphenidate for 6 months. However, polysomnography findings showed significantly increased total sleep time and reduced periodic limb movement index, while the pediatric sleep questionnaire findings showed significant reduction in snoring. This suggests that while methylphenidate treatment may improve certain non-respiratory aspects of sleep in children with ADHD, it does not have a positive effect on sleep disordered breathing (Chin et al., 2018).

Discussion

Our systematic review and meta-analysis of 11 studies and found that close to half of children with ADHD have concomitant OSA, suggestive of a strong association between the two conditions. Descriptive analysis demonstrated that adenotonsillectomy consistently leads to improvement in both OSA and ADHD symptoms, and may be considered as an option before methylphenidate or as a complement to

existing pharmacological therapy in improving ADHD symptoms in children with both conditions.

In establishing the relationship between OSA and ADHD, it has been hypothesized that OSA is one of the five sleep phenotypes associated with ADHD, with the other phenotypes comprising delayed sleep onset insomnia, restless legs syndrome, narcoleptic and sleep EEG epileptiform discharges (Miano et al., 2019). With regards to biochemical changes, a previous study found that the systemic inflammatory response caused by OSA led to increased levels of inflammatory mediators like C-reactive protein and interleukin-6, which correlate with sleep disruption and hypoxia (Tauman et al., 2004). Children with OSA also demonstrated evidence of decreased arterial blood oxygen saturation and brain hypoxemia (Kim et al., 2018). Such a phenomenon then limits the restorative benefits of sleep, disrupting cellular and chemical balances, eventually leading to prefrontal cortical dysfunction which may manifest as features of overactivity and impulsivity (Beebe & Gozal, 2002). Furthermore, sleep deprivation and fragmented sleep due to OSA can result in exacerbation of ADHD symptoms. This may explain why OSA and ADHD are strongly linked and may present concurrently in childhood.

Nonetheless, we acknowledge that there exists heterogeneity in our analysis of the prevalence rates of ADHD in children with OSA. Notably, Nguyen et al. reported a high prevalence of 43.1% of ADHD in children with OSA. Current epidemiological literature suggests a prevalence of 20% to 30% of ADHD in children with OSA (Nguyen-Ngoc-Quynh et al., 2022; Urbano et al., 2021), which is similar to figures provided by Dillon et al. This may be attributed the fact that Nguyen et al. recruited subjects who were all diagnosed with asthma, regardless of whether they had OSA. A previous systematic review and meta-analysis by Kaas et al. demonstrated a positive association between asthma and ADHD after adjusting for relevant confounders (Kaas et al., 2021). A large-scale genome-wide cross-trait analysis by Zhu et al. also found a significant genetic correlation between ADHD and asthma with seven jointly associated loci (Zhu et al., 2019). Since children with asthma are more likely to have ADHD, this could have led to significant confounding and hence a higher prevalence of ADHD among the children with OSA in this study.

Adenotonsillectomy may improve ADHD symptoms in children by improving overall sleep architecture and providing better quality of sleep which is critical in promoting optimal neurocognitive development in early adolescence, in areas including attention, reward, emotional processing and regulation, memory, and executive control (Brooks et al., 2022). In fact, children with habitual snoring but without clinically significant hypoxia as measured by pulse oximetry have been shown to have poorer academic performance than those who do not snore (Brockmann et al., 2015). A prospective, nonrandomized study by Wei et al. showed that in 117 children with sleep disordered breathing who underwent adenotonsillectomy, correlations between sleep and behavior scores were statistically significant before and after surgery. Not only did both behavior and sleep improve independently before and after adenotonsillectomy for sleep disordered breathing in this group of patients, they also improved in correlation with each other (Wei et al., 2007). As such, the associations between poor sleep and neurocognitive, behavioral issues in children are evident, and adenotonsillectomy aims to restore proper sleep quality.

Our findings are consistent with the Childhood Adenotonsillectomy Trial (CHAT) in 2013. The CHAT trial was a multicenter, single-blind, randomized study investigating the on neurocognitive and behavioral outcomes in children aged 5 to 9 years old with OSA who were subjected to early adenotonsillectomy versus watchful waiting. The study demonstrated significantly greater behavioral improvements post-surgery, as reported by caregivers and teachers using the Conners' Rating Scale. In addition, modest improvement for the adenotonsillectomy group was noted compared to the watchful waiting group in the attention and executive-function score on the Developmental Neuropsychological Assessment ($p = .06$; Redline et al., 2011). For the purposes of our study, the CHAT trial was excluded as participants did not have a formal diagnosis of ADHD; instead, general cognitive and behavioral outcomes were analyzed.

Furthermore, several studies examining children with sleep-disordered breathing and ADHD have similarly shown substantial improvement in ADHD symptoms following adenotonsillectomy. Ahmadi et al. demonstrated that in 59 children with ADHD, adenotonsillectomy led to statistically significant change in ADHD rating scale score in both domains, in 1 month and 3 months follow-up period ($p = .001$; Ahmadi et al., 2016). Another study by Amiri et al. showed similar findings with adenotonsillectomy leading to significant reduction in the severity of ADHD symptoms (oppositional behavior, cognitive disorders, inattention, hyperactivity and ADHD index) at 3- and 6-month postoperative intervals (Amiri et al., 2015). These studies were not included in our current systematic review and meta-analysis as children did not have a formal diagnosis of pediatric OSA. Nonetheless, it is evident from current literature that adenotonsillectomy improves sleep quality which leads to enhanced cognitive function, reduced hyperactive behaviors and improved attention consistent across different rating scales.

It is interesting to note that adenotonsillectomy may potentially give rise to greater improvement in ADHD symptoms than methylphenidate. While both interventions can ameliorate inattentiveness, hyperactivity, and impulsivity, they target distinct underlying mechanisms: methylphenidate addresses neurochemical dysregulation within dopaminergic

and noradrenergic pathways and modulates attention networks, whereas adenotonsillectomy alleviates airway obstruction during sleep, a potentially reversible cause of neurobehavioral dysfunction. It is known that OSA can cause ADHD-like symptoms or worsen ADHD symptoms (Youssef et al., 2011). This overlap in behavioral manifestations between OSA-related sleep fragmentation and ADHD often leads to diagnostic ambiguity. The symptoms of children initially identified to have ADHD may be secondary to underlying OSA, which could explain the substantial improvement in ADHD symptoms following adenotonsillectomy. As such, physicians should consider sleep evaluation concurrently with ADHD work-up in this group, especially in the presence of other supporting symptoms, which can help to reduce unnecessary long-term stimulant use and yield more sustained cognitive and behavioral benefits. Future studies providing a direct, head-to-head comparison between the two therapeutic modalities would be valuable and help to pave the way in formulating individualized, pathophysiology-driven treatment approaches for ADHD symptoms.

While our study has found that adenotonsillectomy improves ADHD symptoms, it would be worthwhile to evaluate if the degree of adenotonsillar hypertrophy affects subsequent outcomes. In Huang et al. 56% of study participants who underwent adenotonsillectomy had grade 1 or 2 tonsils, and 20% did not have adenoid enlargement as evaluated on naso endoscopy and lateral neck radiographs. All other studies did not report data on the degree of adenotonsillar hypertrophy and did not conduct any subgroup analysis to adjust for the above.

The strong association between the two conditions in the pediatric population should prompt physicians to assess and evaluate for OSA in ADHD, which can then guide subsequent treatment strategies. Both OSA and ADHD in children are associated with long-term health consequences, including metabolic syndrome, cardiovascular morbidity, neurocognitive deficits and depression in OSA, (Blechner & Williamson, 2016) as well poor somatic and psychiatric outcomes in ADHD (French et al., 2024). While central nervous system stimulants are typically considered to be first line treatment of ADHD, it is meaningful to appreciate that adenotonsillectomy may improve ADHD symptoms when co-diagnosed with OSA together. Moreover, it is widely known that methylphenidate can be associated with side-effects including insomnia, irritability and anxiety, reduced appetite, hypertension, tachycardia, tics, growth retardation and even suicidal ideation (Verghese et al., 2024). As such, the potential of adenotonsillectomy in improving ADHD symptoms may shape the future of ADHD treatment for children with co-occurring OSA.

Strengths and Limitations

To our knowledge, this study is the first systematic review and meta-analysis that describes the bidirectional relationship of PSG-diagnosed OSA and ADHD diagnosed in the same setting in a pediatric population. While a previous systematic review by Urbano et. al similarly studied the association between sleep-disordered breathing and ADHD, our study differs in evaluating the bidirectional relationship of pediatric OSA and ADHD specifically with objective clinical diagnosis via polysomnography and DSM criteria respectively. It is widely known that children with sleep disturbances often display behavioral patterns that resemble some features of ADHD, (Horiuchi et al., 2014) but may not fulfill its full criteria, limiting the strength of conclusions drawn from previous review papers.

Another strength of our study includes the use of a rigorous protocol and structured methodology to systematically search and grade the included studies based on international guidelines. All included studies were of good quality and had a sizeable sample size, and the longitudinal studies had sufficient follow-up duration. Additionally, none of our included studies had high risk of bias. Moreover, we adopted a well-defined scope of analysis and only included studies which utilized gold standard, validated means of objective diagnosis for OSA and ADHD. We intentionally excluded papers examining general sleep-disordered breathing disorders, which may encompass central sleep apnea, complex sleep apnea and hypoventilation syndromes with different underlying pathophysiology, as well as primary snoring disorders. As such, our study presents a focused and precise analysis on the complex relationship of OSA and ADHD while avoiding confounding factors.

Nonetheless, our study must be interpreted in the context of known and potential limitations. Firstly, there were relatively few studies investigating pediatric OSA and ADHD. Only a handful of studies used formal polysomnography to diagnose pediatric OSA, likely due to its time-consuming and cost ineffective nature, as well as difficulty in administering the test on young children. Instead, majority of studies relied on parent questionnaires or screening tools for OSA symptoms alone, such as use of the Pediatric Sleep Questionnaire, which can be subjective. Owing to limited number of studies, we were unable to adjust for confounders including obesity, allergic rhinitis, asthma, depression, learning disabilities, all of which can affect both OSA and ADHD in children (Nguyen-Ngoc-Quynh et al., 2022; Prajsuchanai et al., 2022). Furthermore, there was heterogeneity among studies related to baseline demographic characteristics of participants and clinical subtype of ADHD (inattentive vs. hyperactive vs. combined). Different studies also used various clinical checklists and rating scales for assessment of ADHD. Inclusion of a greater number of studies would allow us to perform additional analyses to adjust for these confounders and draw more robust conclusions on the relationship and optimal treatment approaches for both OSA and ADHD.

Moreover, while majority of studies have demonstrated clear short-term benefits of adenotonsillectomy on ADHD

symptoms in pediatric OSA patients, few have evaluated the long-term benefits of adenotonsillectomy. As OSA has been known to recur in some children post-adenotonsillectomy, it would be useful to determine the long-term behavioral changes in this group of children.

Future Studies

Future studies should focus on addressing current gaps in understanding causality, mechanisms, and treatment outcomes. Longitudinal cohort studies with longer follow-up durations would be helpful in determining whether early identification and treatment of OSA can reduce the onset or severity of ADHD symptoms over time. Neuroimaging and biomarker-based research may help uncover shared neurobiological pathways, such as dysregulation in dopamine or inflammatory markers. This can lead to development of novel pharmacological therapies. For example, Meliante et al. previously studied the potential role of antioxidant therapy with vitamin C or N-acetylcysteine in reducing oxidative stress produced during episodes of intermittent hypoxia in OSA (Meliante et al., 2023).

It would also be interesting to explore how different ADHD subtypes can possibly relate to specific patterns of sleep disruption such as arousals or rapid eye movement (REM) suppression. This is especially in light of promising developments in wearable sleep technology in the consumer market today, allowing for convenient yet accurate means for diagnosis.

Lastly, future studies should investigate if other OSA treatment including CPAP therapy and orthodontic devices can improve ADHD symptoms in children. Hobzova et al. found that in the adult population, CPAP therapy demonstrated significant improvement in attention, working memory, and depressive symptoms in OSA patients, (Hobzova et al., 2017) while Escobar et al. previously hypothesized the use of oral appliance therapy for OSA treatment may reduce the symptoms of ADHD (Escobar et al., 2021).

Conclusion

OSA and ADHD are closely related conditions of childhood with overlapping symptomatology. Adenotonsillectomy consistently leads to statistically significant improvement in both ADHD and OSA symptoms and may be considered as an option before or as complementary therapy to methylphenidate in improving symptoms of inattention and hyperactivity in children with both conditions. Moreover, children with ADHD symptoms would benefit from sleep evaluation concurrent with ADHD work-up. A greater number of large-scale, robust studies are required to further evaluate the treatment outcomes between OSA and ADHD.

ORCID iD

Claire Jing-Wen Tan https://orcid.org/0000-0003-4529-2978

Ethical Considerations

This research paper is an original work conducted and written by all authors. All sources of information and ideas that are not our own have been appropriately cited. This work has not been submitted, in whole or in part, for publication or academic credit elsewhere.

Consent to Participate

Not applicable as no study participants are involved in this study.

Authors Contributions

BLHW, CJWT, BYSY, ANCW, TOH, TST, and ETGH designed the study and developed the study protocol and tools. BLHW, CJWT, NNNE and SLCY were responsible for data collection. BLHW, CJWT and BYSY analyzed data and wrote the manuscript. All authors read and approved the final manuscript.

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Data Availability Statement

Data used for this study can be accessed upon request from the Principal Investigator (Dr. Ellen Tay Ghim Hoon) at ellen.tay.g.h@singhealth.com.sg.

Supplemental Material

Supplemental material for this article is available online.

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Author Biographies

Bryan Hao Wei Leow is a Medical Officer in Singapore General Hospital Otolaryngology Department.

Claire Jing-Wen Tan is a Junior Resident in Singapore General Hospital Otolaryngology Department.

Brian Sheng Yep Yeo is a House Officer in Singapore General Hospital Otolaryngology Department.

Noelle Na En Ng is a medical student in National University of Singapore.

Seraphina Cheng Yong Low is a medical student in National University of Singapore.

Adele Chin Wei Ng is an Associate Consultant Otorhinolaryngologist at Singapore General Hospital and Singhealth Duke-NUS Sleep Centre, with special interest in treatment of snoring and obstructive sleep apnea related conditions.

Oon Hoe Teo is a Senior Consultant at KK Women's and Children's Hospital, Singapore and Singhealth Duke-NUS Sleep Centre. He has a keen clinical interest in paediatric respiratory medicine and treatment of asthma in children.

Song Tar Toh is a Senior Consultant Otorhinolaryngologist at Singapore General Hospital and Singhealth Duke-NUS Sleep Centre, with special interest in treatment of snoring and obstructive sleep apnea related conditions. He has extensive experience in transoral robotic surgery and hypoglossal nerve stimulation implant surgery.

Dr Ellen Ghim Hoon Tay is a Senior Consultant in the Department of Child Development at KK Women's and Children's Hospital (KKH), with an interest in caring for children with developmental, behavioural, and learning needs.